Objectives

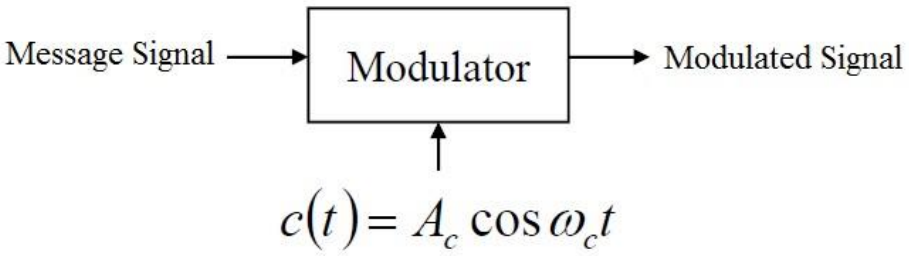
A) To design transmitter and receiver for DSB-SC modulation.

B) To calculate power efficiency (power of message/ total power)

C) To calculate bandwidth of the message signal.

D) To calculate bandwidth of modulated signal.

In communication, modulation is the process of varying one or more properties of a highfrequency periodic waveform, called the carrier signal, with a modulating signal which typically contains information to be transmitted, called the baseband (message signal). A modulator is used to modify a carrier signal's parameter (frequency, amplitude, phase, etc.) in accordance to the baseband signal.

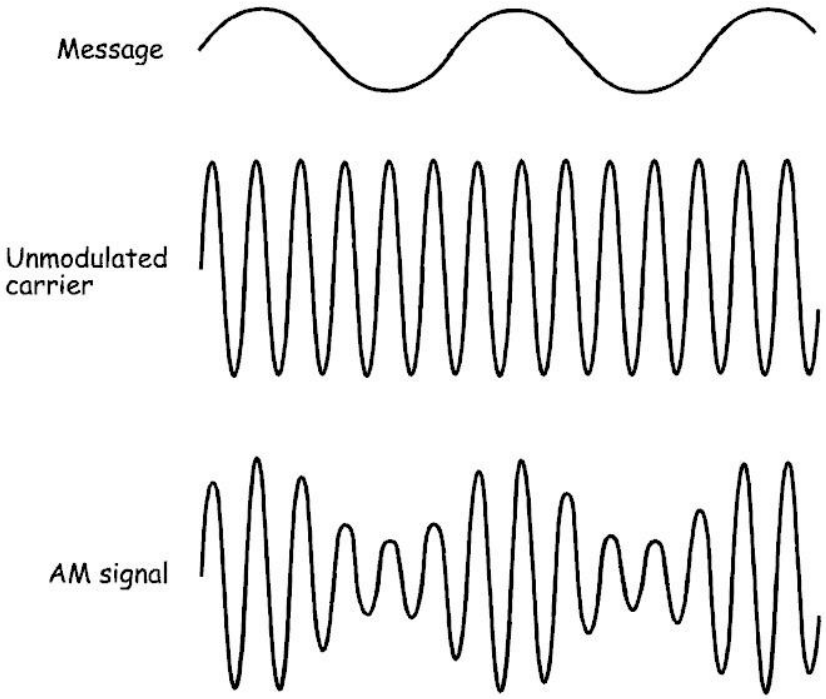


Modulation is necessary in the transmission of data, such as an audio signal, over a long range. Baseband signals are inherently limited in range because their frequency range is typically in a lower range and their power makes the signal unsuitable for transmission. Modulation gives several advantages, such as reducing the size of transmitting antenna, better usages of limited bandwidth, and much more.

I. Theory

In an amplitude modulation (AM) communications system, the message signal is used to vary the amplitude of a pure cosine wave called the carrier. The carrier usually has a frequency (f_c) that is much higher than the message's frequency. Figure below shows a simple message signal and an unmodulated carrier. It also shows the result of amplitude modulating the carrier

with the message. Notice that the modulated carrier's amplitude varies above and below its unmodulated amplitude.



Double Sideband Supressed Carrier (DSB-SC) Modulation:

Let's denote the baseband message signal with $m\ t( )$ and the carrier signal with $c\ t( ) = A_c \cos 2(\square f\ t_c)$ where A_c is amplitude and f_c is carrier frequency. The DSB-SC modulated signal is given as

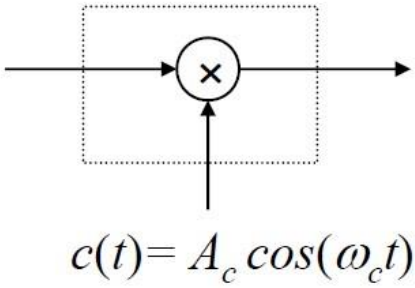
$$u\ t( ) = m\ t\ c\ t( )\ ( ) = m\ t\ A( )\ _c \cos 2(\square f\ t_c) \quad (1)$$

The spectrum of the modulated signal is

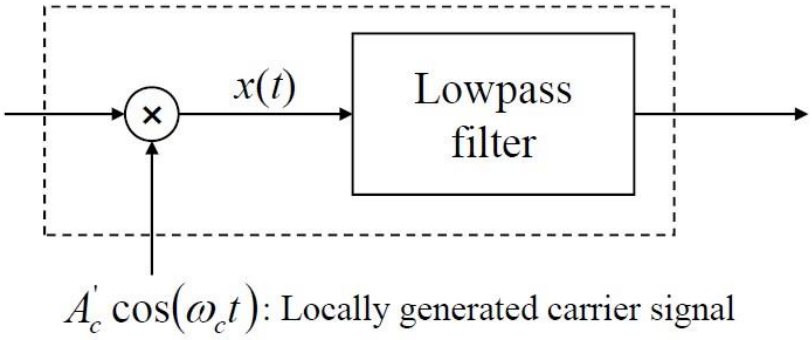
$$U f( ) = \frac{A^c M f(- f_c)}{} \quad \frac{A^c M f(+ f_c)}{}$$

2 2 where $M(f)$ is spectrum of message signal $m(t)$, and $U(f)$ is shifted versions of $M(f)$ into f_c and $-f_c$ frequencies.

Modulation: DSC-SB modulation is multiplication of message signal $m(t)$ and $c(t)$.



Demodulation: For DSB-SC, we use coherent demodulation as shown in figure below.



Note: Use lab computers and one USRP!

Experiment 5.1 (Transmitter, DSB-SC with USRP)

In previous experiments we could record our voice, save it into a queue and playback it using LabVIEW. And also we learned about DSB-SC modulation. Now, it is time to transmit our voice using USRPs with DSB-SC modulation from one computer to another one and listen to it. In order to do this, after recording the sound, we need to take the data out of the queue, resample it to sampling rate of **USRP (200 kSa/s)** and apply DSB-SC or SSB modulation. Then the modulated signal is entered to the USRP for transmission. At the receiver side, we need to demodulate the received signal using coherent demodulation and filter it using a lowpass filter.

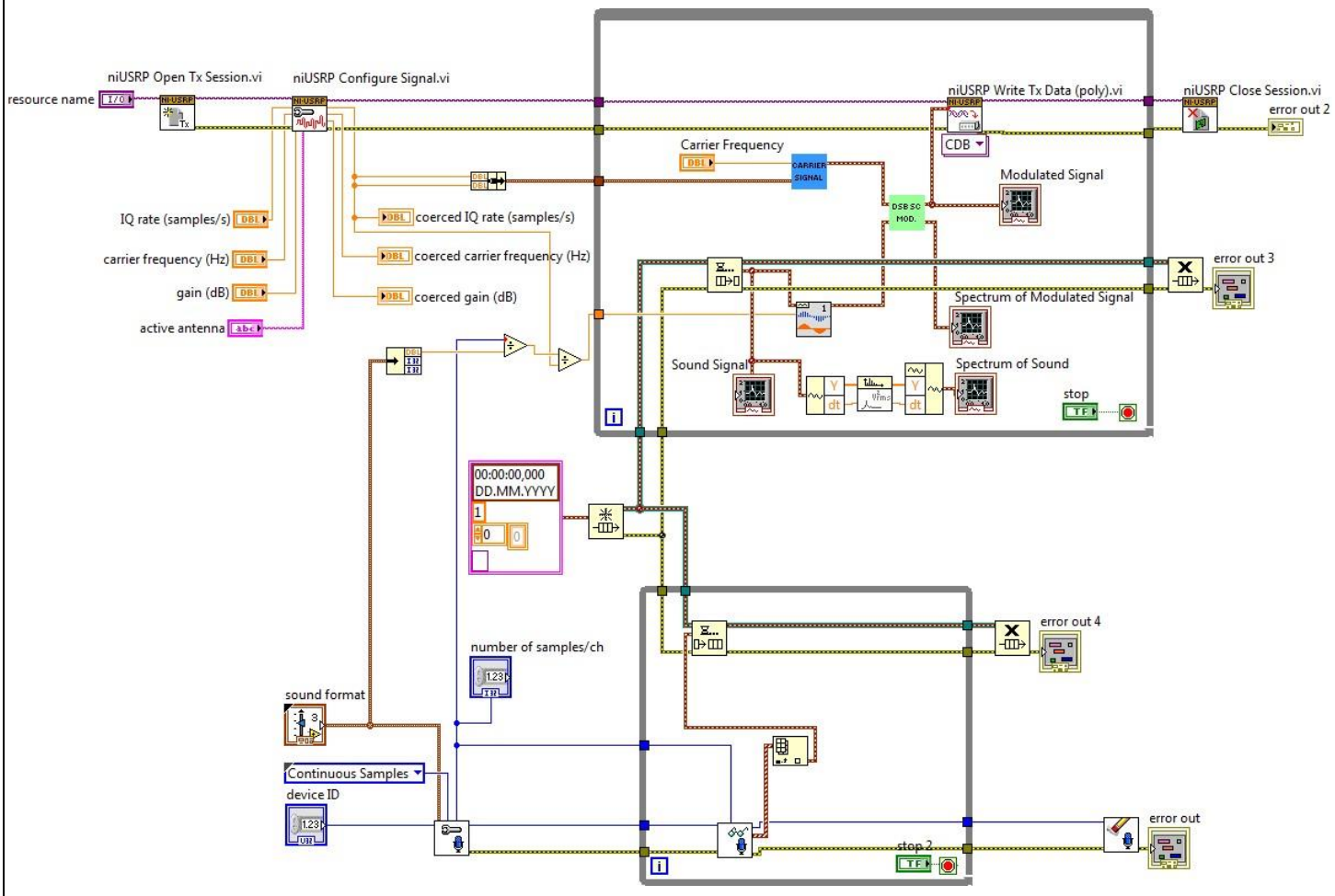
Then we resample the signal to sampling rate of sound card (44100 Sa/s) and play it. The voice heard is with a delay (because of processing times), is noisy (because of noise of channel). In this experiment we are going to record our voice and save it in a queue same as Experiment 3.

Then we use DSB-SC modulation (SubVI of prelab) to transmit the signal using USRP. In this experiment we will use USRP to build an analog communication system. USRP can be easily configured using LabVIEW software and the PC controls the USRP through the gigabit Ethernet cable connecting.

In order to build the transmitter follow the below procedure:

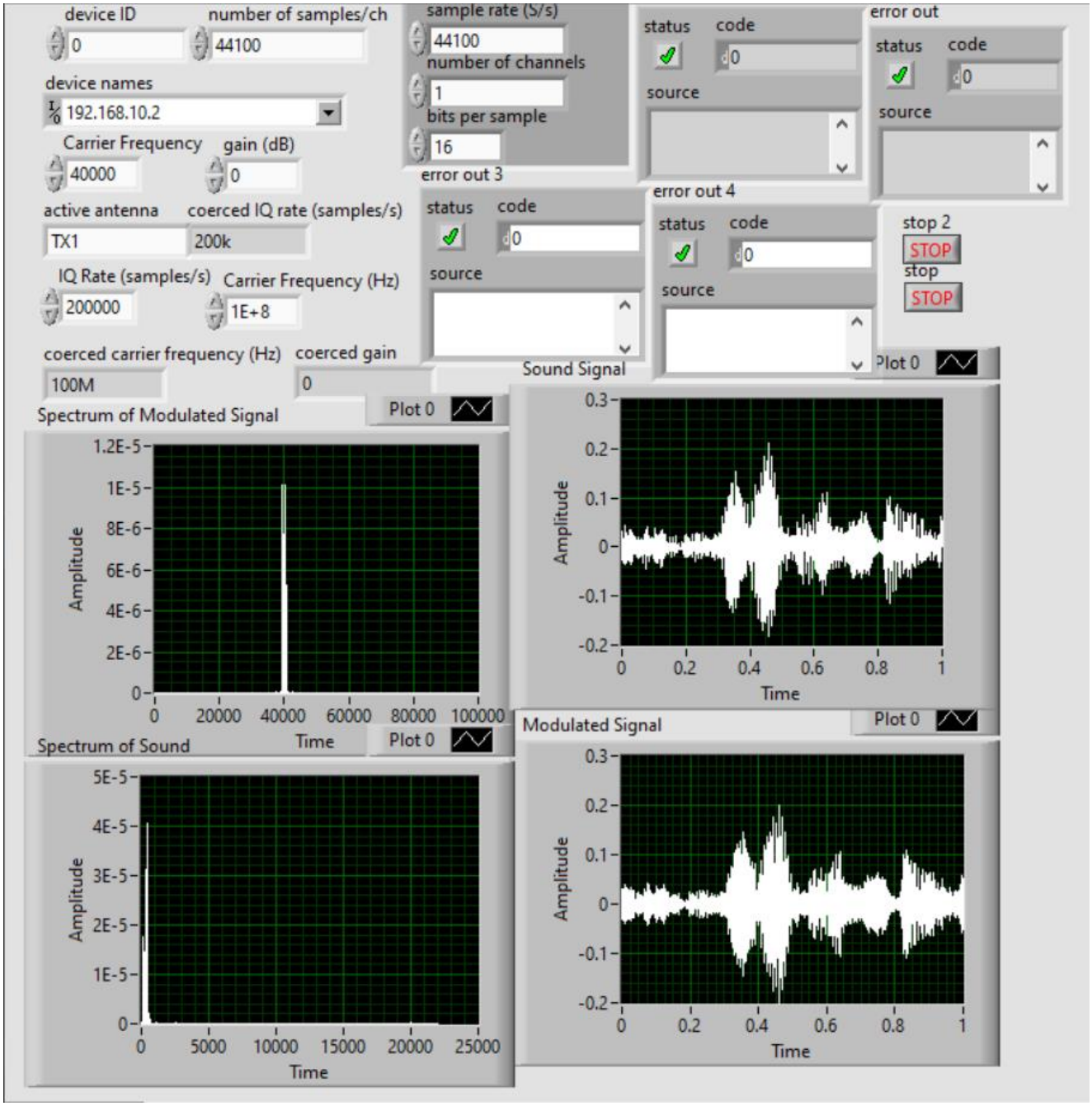
1. Create a new VI.
2. First of all, we record our voice and save it in a queue (the bottom part of the following figure). For this reason we use the VI file of Experiment 3.1 Copy the recording part of Experiment 3.1 VI file (the top part of the block diagram) and modify it based on the following figure. Set the **number of samples/ch = 44100**.
3. For the top part of the following figure we use the USRP function blocks in **Instrument I/O => Instrument Drivers => NI-USRP => Tx.**
4. Add the following blocks in order: niUSRP Open Tx Session, niUSRP Configure Signal, niUSRP Write Tx Data, niUSRP Close Session.
5. Create the control "resource name" for "niUSRP Open Tx Session" block. Set it to **192.168.10.2** the IP address of USRPs.
6. Also create the following controls and indicators for "niUSRP Configure Signal": IQ rate (samples/s) = 200 k carrier frequency (Hz) = 100 MHz gain (dB) = 0 active antenna = TX1
7. In order to provide the data for USRP, take the element port of Dequeue block and resample it to 200 kSa/s (sampling rate of USRP) using Resample Waveform VI.
8. Also use a graph to show the sound signal (message signal). Same as previous experiment show the spectrum of message signal in graph.

9. For carrier signal use your Sub VI “CarrierSignal.vi” and set the parameters. (**carrier frequency = 40 kHz**, **Fs = Coerced IQ rate**, **#s = Coerced IQ rate**)
10. For DSB-SC modulation use your Sub VI “DSBSC Modulation.vi” and connect the carrier signal and message signal to it.
11. Finally, connect the modulated signal to input of “ni USRP Write Tx Dara”.
12. Show the modulated signal and its spectrum using graphs. **(When you finished your block diagram check it with instructor and save it)**



Questions:

1. Make sure the transmitter is working correctly and take a screenshot of the graphs.



2. Congratulations! you just made an AM radio transmitter. We continue to the receiver part in the next lab.